

PROJECT MANAGEMENT PLAN

Division of Work · Sub-Issues · Milestones

Trustworthy Deep Learning for Chest X-ray Disease Detection

5-Member Work Breakdown Structure — everyone hands-on across the full data-science & DNN process

The organising principle

Every member owns ONE deep-learning model end-to-end — not just 'a job'. Each model owner personally runs the full DNN workflow on their model: data → train (frozen + fine-tune) → calibrate → explain → robustness test → efficiency benchmark.

The three CNNs are split across three members; the Vision Transformer goes to a fourth; the fifth trains the U-Net segmentation model and owns evaluation. So all five are in the DNN process.

Everyone ALSO owns one shared 'standard' (data / efficiency / explainability / calibration+robustness / reproducibility) so all results are directly comparable, plus paper sections.

Contents

1. How to Use This Plan

This document is the human-readable management plan; the accompanying Excel workbook ('Project_Work_Breakdown_Tracker.xlsx') is the live tracker your team updates daily. Use them together:

- **This document** — for the team model, per-person responsibilities, milestones, dependencies and coordination rhythm. Read once at kick-off.
- **Excel 'Master Task List'** — single source of truth. Each task has an owner and a Status dropdown (Not started / In progress / Blocked / Done). The Dashboard tab auto-counts progress.
- **Excel per-member boards (M1–M5)** — each person's focused list of owned + supporting tasks.

2. Team Model at a Glance

The team is five people. To guarantee everyone gains real deep-learning experience, we assign models rather than phases — nobody is 'only the writer' or 'only the data person'. Writing and data work are distributed on top of a model each person trains.

Code	Member & role	DNN model owned	Shared lead role	Paper sections
M1	ResNet50 & Reproducibility/Lit Lead	ResNet50 (CNN)	Repo, experiment tracking, seeds, references	Intro, Related Work, Gap, Conclusion, Abstract
M2	DenseNet121 & Data Pipeline Lead	DenseNet121 (CNN)	Preprocessing, augmentation, split, dataloaders	Methodology
M3	EfficientNet-B0 & Efficiency Lead	EfficientNet-B0 (CNN)	Efficiency harness, Pareto, compute setup	Slides
M4	ViT-Base & Explainability Lead	ViT-Base (Transformer)	XAI standard + faithfulness score, figures	Discussion
M5	Segmentation/Eval & Paper Lead	U-Net lung segmentation	Calibration + robustness protocol, external data, stats	Results, final assembly

Why this split works. The 3 CNNs go to M1/M2/M3 (your requirement), the Transformer to M4, and M5 trains a U-Net — so all five train a neural network. Each classification-model owner then re-uses shared standard modules (built by the leads) to run calibration, explainability, robustness and efficiency on their own model, giving each person the complete end-to-end DNN experience while keeping every result comparable.

3. Main Milestones & Gates

Five gates structure the month. A gate must be signed off before dependent work proceeds — this prevents the classic failure of racing ahead on an unfair or broken pipeline.

Gate	Due	Name	Must be true to pass	Sign-off
M0	End W1	Foundations Ready	Shared data pipeline frozen & reproducible; baseline (DenseNet121) trains end-to-end; external data prepared; lit-review drafted.	M2 / M1
M1	End W2	All Models Trained	All 4 classification models trained (frozen + fine-tuned) with accuracy tables; U-Net segmentation ready.	Owners / M5
M2	Mid W3	Trust Experiments Done	Calibration + explainability + faithfulness complete for all 4 models.	M4 / M5
M3	End W3	Robustness & Efficiency Done	External OOD test + efficiency benchmark for all models; segmentation-ablation done.	M5 / M3
M4	End W4	Camera-Ready	Consolidated analysis, Pareto, full paper draft, slides; reproducibility rerun passes.	M1 / M5

Gate	Due	Name	Must be true to pass	Sign-off
The single rule that protects the whole project Freeze one identical training + preprocessing pipeline at Gate M0 and make every model use it. If models are tuned differently the comparison is not publishable — fairness is the entire point of the study.				

4. Weekly Cadence (who does what, each week)

Member	Week 1	Week 2	Week 3	Week 4
M1	Repo, tracking, seeds; start lit review	Train ResNet50 (frozen+FT)	ResNet50 calibration, XAI, external test	Consolidate w/ M5; Intro/Conclusion/Abstract
M2	Build shared data pipeline; baseline gate	Train DenseNet121; support seg data	DenseNet calibration, XAI, external test	Methodology section
M3	Compute setup; env pinning	Train EfficientNet-B0	EffNet calibration/XAI/OOD; efficiency harness + benchmark	Pareto analysis; slides
M4	Scope XAI standard	Train ViT-Base	Build XAI module; ViT calibration/rollout; all faithfulness	Figures; Discussion
M5	Shared config; external data; train template	Train U-Net; masks; calibration standard	Robustness protocol; segmentation ablation; run all OOD	Results; stats; reproducibility rerun; assembly

5. Per-Person Breakdown (sub-issues)

Member 1 — ResNet50 & Reproducibility / Literature Lead

DNN model owned	ResNet50 (CNN)
Shared lead role	Repo, experiment tracking, seeds, references
Paper sections	Introduction, Related Work, Research Gap, Conclusion, Abstract

Sub-issues you own:

- Set up GitHub repo, branch strategy and folder layout (T01).
- Configure experiment tracking (W&B/TensorBoard) and the fixed-seed reproducibility standard (T02).
- Train ResNet50 — Phase 1 frozen, then Phase 2 fine-tune + hyperparameter tuning (T12–T13).
- Run ResNet50 calibration, explainability + faithfulness, and the external OOD test (T24, T29, T34).
- Write Introduction, Related Work and the Research-Gap section from Week 1; manage references in Zotero (T45, T51).
- Write Conclusion + Abstract in Week 4 (T49).

Also supporting: Repo/tracking for all members; consolidation review with M5.

Member 2 — DenseNet121 & Data Pipeline Lead

DNN model owned	DenseNet121 (CNN)
Shared lead role	Preprocessing, augmentation, split, dataloaders (shared)
Paper sections	Methodology

Sub-issues you own:

- Download the COVID-19 Radiography dataset; integrity check and class-count report (T05).
- Build the shared preprocessing, augmentation, split and DataLoader modules everyone imports (T06–T09).
- Run the baseline-gate model end-to-end to validate the pipeline before the team scales up (T11).
- Train DenseNet121 — frozen then fine-tune + tuning (T14–T15).
- Run DenseNet121 calibration, explainability + faithfulness, external OOD test (T25, T30, T35).
- Own the Methodology section of the paper (T46).

Also supporting: Segmentation data with M5; supports most Week-1 setup tasks.

Member 3 — EfficientNet-B0 & Efficiency Lead

DNN model owned	EfficientNet-B0 (CNN)
Shared lead role	Efficiency benchmark harness, Pareto, compute setup
Paper sections	Slides

Sub-issues you own:

- Set up shared compute (Colab/Kaggle GPU) and pin the environment (T03).
- Train EfficientNet-B0 — frozen then fine-tune + tuning (T16–T17).
- Run EfficientNet-B0 calibration, explainability + faithfulness, external OOD test (T26, T31, T36).
- Build the efficiency harness (params, FLOPs, latency CPU/GPU, GPU memory) and benchmark all 4 models (T39–T40).
- Lead the Pareto trade-off analysis (accuracy vs trust vs efficiency) (T43).
- Build the presentation slides (T50).

Also supporting: Preprocessing help; figures with M4.

Member 4 — ViT-Base & Explainability Lead

DNN model owned	ViT-Base (Vision Transformer)
Shared lead role	XAI standard (Grad-CAM / attention rollout / SHAP / faithfulness), figures
Paper sections	Discussion

Sub-issues you own:

- Train ViT-Base — frozen then fine-tune, with DeiT/ViT-Small fallback if unstable on the small dataset (T18–T19).
- Build the shared explainability module: Grad-CAM (CNNs), attention rollout (ViT), SHAP, and the lung-localisation faithfulness score (T28).
- Run ViT calibration and attention-rollout faithfulness (T27, T32).
- Run ViT external OOD test (T37).
- Generate all figures — learning curves, heatmaps, reliability diagrams, Pareto (T44).
- Own the Discussion + model-selection guidance section (T48).

Also supporting: Explainability standard used by every model owner; figures with M3.

Member 5 — Segmentation / Evaluation & Paper Lead

DNN model owned	U-Net lung segmentation
Shared lead role	Calibration + robustness protocol, external data, statistics
Paper sections	Results, final assembly

Sub-issues you own:

- Define the shared config (image size, splits, metrics schema, checkpoint format) and the training-loop template (T04, T20).
- Download and convert the external RSNA set (DICOM→PNG) and map labels (T10).
- Train the U-Net lung-segmentation model and generate masks for primary + external sets (T21–T22).
- Build the calibration standard (ECE/Brier/reliability/temperature scaling) and the external-test/OOD protocol (T23, T33).
- Run the full-image vs lung-segmented shortcut-mitigation experiment (T38).
- Consolidate all metrics, run statistical comparison, own the Results section, and lead the final reproducibility rerun and assembly (T41–T42, T47, T52).

Also supporting: Calibration & robustness standards used by every model owner; co-owns the manuscript with M1.

6. Shared Standards (so results stay comparable)

Each shared standard is built once by its lead, then used identically by every model owner. This is what turns five people's separate models into one rigorous, comparable benchmark.

Shared standard	Built by	Used by	Purpose
Data pipeline (preprocess / augment / split / loaders)	M2	All owners	Identical inputs for every model — fair comparison.
Training-loop template (frozen → fine-tune)	M5	All owners	Same optimiser/schedule/seed across models.
Calibration module (ECE, Brier, temp scaling)	M5	All owners	Comparable reliability numbers.
Explainability module (Grad-CAM / rollout / SHAP / faithfulness)	M4	All owners	Comparable, quantified explanations.
External-test / OOD protocol	M5	All owners	Comparable robustness (generalisation gap).
Efficiency harness (params/FLOPs/latency/memory)	M3	All owners	Comparable deployment metrics.
Reproducibility (repo, seeds, tracking)	M1	All owners	Anyone can re-run any result.

7. Critical Dependencies & Sequencing

- **Nothing trains until the data pipeline (M2) and training template (M5) are frozen at Gate M0.** Model owners are blocked until then — so Week 1 is all-hands on the pipeline and baseline.
- **Explainability faithfulness (M4's module) needs the segmentation masks (M5).** M5 must deliver masks by early Week 3.
- **Calibration, XAI and OOD tests per model** each depend on that model being fully trained (Gate M1).
- **Consolidation, Pareto, Results and slides (Week 4)** depend on every model's metrics landing in the master results table (T41).
- **Writing is continuous, not last-minute.** M1 drafts Intro/Related Work from Week 1; M2 drafts Methodology from Week 2.

8. Coordination Rhythm (management tips)

- **Daily 10-minute stand-up:** each member says what they did, will do, and any blocker. Update Status in the Excel tracker before stand-up.
- **Gate reviews:** a short sign-off meeting at each milestone (end W1, end W2, mid W3, end W3, end W4). Do not start dependent work until the gate owner signs off.
- **Code review via pull requests:** no direct commits to main; at least one peer approves. Shared modules get extra scrutiny.
- **Results hygiene:** every metric is logged to results/ in the agreed schema so the Results author (M5) never chases numbers.
- **Risk buffer:** if time is short, cut the optional NIH check and SHAP first; protect the five-axis core (accuracy, calibration, faithfulness, robustness, efficiency).